

Alphanumeric pagers have LCD screens that display text and numbers, work on radio frequencies and are one-way devices, meaning they can only receive messages



5

During manufacturing, or during their interception, the pager batteries were likely laced with the explosive Pentaerythritol tetranitrate (PETN) and connected to a chip





On September 17, around 3.30 pm Beirut time, thousands of pagers used by Hezbollah operatives exploded in markets, homes and vehicles, killing 11 and injuring thousands



9

Hunny Bakshi (retd), former commander of an intelligence unit created after the 26/11 Mumbai attacks.

Lt Gen. Prakash Katoch (retd), former Director General of Information Systems of the Indian Army, says, "China has the capability to carry out cyber kinetic attacks anywhere, as it dominates the electronic market globally. In India, we don't even have a 100 per cent check of imported items for embedded malware. And even surveillance devices like CCTVs (including in sensitive locations) are Chinese-origin."

Indeed, experts have raised an alarm over the unchecked proliferation of Chinese-made closed circuit television (CCTV) cameras installed in critical government institutions in India like the naval establishments and the Defence Research and Development Organisation (DRDO). These devices potentially allow remote surveillance and data breaches that Beijing could exploit. The threat is real, for in the past few years the US, UK and Australia have all banned the use of Chinese CCTVs. Now, the Union home ministry has set up a wing of intelligence officials to monitor Chinese firms' engagement with Indian projects. Additionally, the government is working to phase out Chinese equipment within the national security apparatus.

Experts are also concerned that Chinese manufacturers dominate the global supply of Cellular IoT Modules (CIMs), which enable the working of IoT by extending connectivity to everyday physical devices via the cellular network (4G/ 5G). CIMs are embedded in a wide range of devices, from personal gadgets, CCTVs and drones to production lines and power grids. CIMs can receive firmware updates remotely without the end user's awareness. These could carry malware, enabling the Chinese to manipulate devices, shut them down, or steal data. A cyber kinetic attack can also not be discounted. How does India deal with this? Experts say the government must ensure strict testing of CIMs supplied by foreign vendors against globally accepted security standards. India should also develop indigenous CIMs for foolproof security.

Particularly vulnerable to cyber kinetic attack is the aviation sector worldwide, which increasingly uses inflight Wi-Fi and fly-by-wire systems—computer-regulated aircraft flight control systems that replace mechanical/ manual flight controls with an electronic interface. An Indian security official working in the aviation sector says the Lebanon attacks have shaken the trust in existing safety protocols. Jaijit Bhattacharya, president of the New Delhi-based Centre for Digital Economy Policy Research (C-DEP), stresses the all-pervasive reach of cyber kinetic warfare—a smart car with built-in explosives that can blow up after cyber surveillance detects a passenger, or drones that can search and kill humans individually. "The possibilities of combining cyber with Artificial Intelligence and kinetics are mindboggling and will have a deep impact on conflicts," says Bhattacharya.

Soon after the Lebanon attacks, the Indian Army started a course for officers on how future wars will be fought in the cyber, space and electromagnetic domains while exploring the impact of AI, machine learning and hypersonics. After the events of September 17-18, fields of conflict have expanded; India must be battle-ready to meet its exigencies. ■